Problem Set 11

Due date: never

This problem set will not be graded. It is for educational and entertainment purposes only!

1. **Black Holes with a Negative Mass:** suppose we wanted to try to make sense of a "fifth region" of Schwarzschild, where $r < 0$. This is not accessible to any observer since $r = 0$ is a genuine singularity – you can't travel through it. Nevertheless, we can still imagine this being a different universe altogether, not accessible from the first one in any way.

   (a) Show that takin $r < 0$ is identical to taking $M < 0$ with a radial direction that takes positive values.

   (b) Is $r = 0$ a genuine singularity in negative mass Schwarzschild? (Hint: this does not require any calculations beyond what is already done in the lecture notes.)

   (c) Is there a region where t is the timelike coordinate?

   (d) Is there a region where r is the timelike coordinate?

   (e) Show that an outgoing null trajectory can always reach infinite future time, no matter what value of r it starts at. (Here by outgoing we mean moving towards larger r).

   (f) Is the spacetime asymptotically flat at $r \to \infty$?

   Negative mass Schwarzschild is an example of a singularity without an event horizon!

2. **Clocks in Schwarzschild:** Consider a static clock at $r > R_s$ (here static means static in Schwarzschild coordinates: fixed r, θ, ϕ).

   (a) How fast does this clock tick compared with a clock at $r = \infty$?

   (b) Suppose that you are standing on the surface of the sun. How fast does your clock tick relative to a far away clock (i.e. at $r = \infty$)? The mass of the sun is $2 \cdot 10^{30} kg$ and the radius at the surface of the sun is $7 \cdot 10^8 m$.

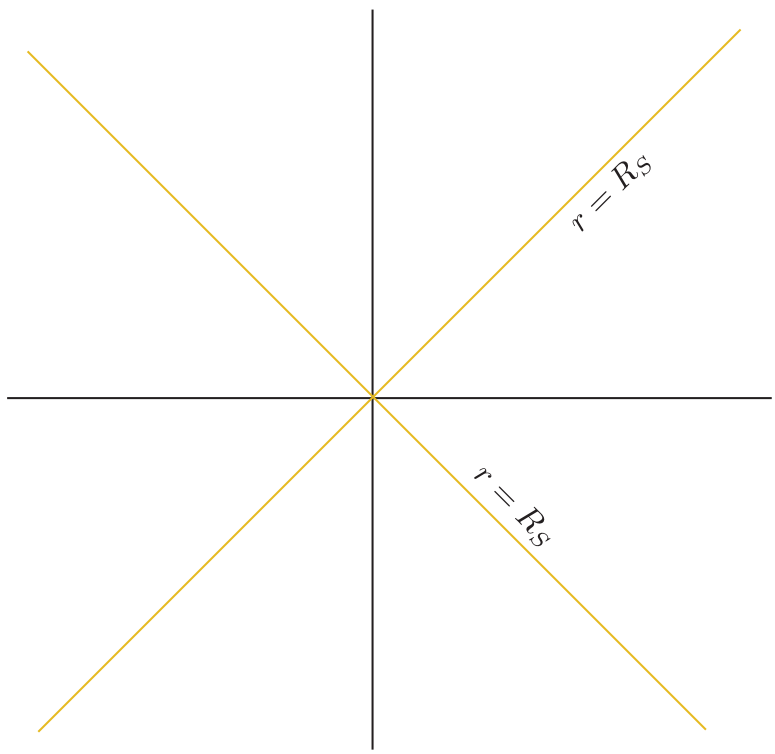
3. **Falling into a Black Hole:** Suppose that you are dropped out of a spaceship outside of a black hole and that you fall in. To answer the questions below, use the fact that the spacetime near the horizon of a black hole is just like the region of flat spacetime near an acceleration horizon.

   (a) On the diagram on the next page, sketch the worldline of a spaceship that stays outside of the horizon and your worldline as you are dropped outside of the spaceship and fall across the horizon.

   (b) Suppose that you send out light rays at regular intervals to signal the rocket. Draw several of these light rays on your diagram.

   (c) The people on the rocket watch the light rays you send out through a telescope. Do they see you age slowly or quickly? When do they see you cross the horizon? If they first sit and watch for a while, is it possible for them to come and rescue you?

   (d) Suppose now that the rocket sends out light rays toward you at regular intervals. Describe what you see if you watch these light rays through a telescope. Do you see people in the rocket age slowly or quickly? Do you see anything special when you cross the horizon?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